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RESEARCH ARTICLE

Advancing Interoperable IoT-Based Access Control Systems: A Unified Security Approach in Diverse Environments

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ABSTRACT The convergence of the Internet of Things (IoT) and the construction industry represents a potent opportunity to fundamentally redefine the processes of constructing and maintaining buildings. The integration of IoT technology has the potential to revolutionize building safety, efficiency, and sustainability, offering a promising avenue for innovation and advancement within the industry. In this work, we propose an innovative design and implementation of an interoperable physical access control IoT-based system that can be easily integrated with existing access control security systems and used in various environments such as offices, homes, and public buildings. Based on our developed and implemented physical access control IoT-based system, we identified several challenges in the current state of access control systems. These challenges include difficulties in integrating different security systems, the absence of interoperable standards, and the need for further research to enhance the scalability and performance of such systems.

INDEX TERMS Internet of Things (IoT), interoperability, physical access control, electronics, smart devices, wireless communication, software design.

I. INTRODUCTION

The safety of occupants and the building itself is paramount in both public and private buildings. Of the various security systems used for building security, door access control is significant. Door access control is a physical security system that restricts access to specific individuals and maintains records of access. With the increasing potential of IoT technology, concerns regarding interoperability and data sharing have become significant issues, especially for physical access control systems relying on IoT technology. In particular, door access control systems that depend on IoT technology face significant challenges in achieving interoperability due to the absence of a uniform standard. Integrating different security systems can be challenging, resulting in inefficiencies and increased costs. It is necessary to develop and adopt

interoperable standards that allow for the seamless integration of different security systems. The adoption of such standards can ensure that physical access control systems based on IoT technology can interoperate and communicate effectively, irrespective of the manufacturer or type of security system.

By addressing interoperability concerns, door access control systems can enhance building security and safeguard occupants and structure by making unauthorized access more difficult. By establishing a centralized control and monitoring, it is easier to manage access permissions, track entry and exit logs, and respond to security incidents in real-time.

In this study, we present an IoT-based access control system that stands at the forefront of technological innovation in security and interoperability. The contributions of this study are outlined as follows:

- Our research's core contribution lies in the development and implementation of a unique system that seamlessly integrates with existing security frameworks while

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introducing cutting-edge features, including the novel use of rolling QR codes for dynamic authentication and enhanced security. Rolling QR codes ensure access is granted only within a specific timeframe, adding an extra layer of security.

- Secondly, the proposed system not only elevates the standard of physical access control but also addresses critical challenges in the IoT realm, such as the lack of interoperable standards and the complexities of integrating diverse security systems.
- Thirdly, the developed system showcases exceptional advancement in software and hardware integration, setting a new benchmark for flexibility, scalability, and efficiency in access control systems. Designing software components that enable robust and efficient communication and data processing, we have enhanced the proposed system's ability to manage diverse security protocols and data formats.
- Finally, the innovative hardware design, characterized by the system's compatibility with both traditional and modern access control mechanisms, exemplifies our commitment to bridging the gap between legacy systems and future technologies.

This paper outlines a blueprint for future developments and a solid foundation for further exploration and enhancement of access control systems. The implications of our work extend beyond the immediate application, providing valuable insights and methodologies that can be applied across various domains where security, interoperability, and technological integration are paramount.

This paper is organized as follows: Section II delves into the existing literature and the current state of access control systems, setting the stage for our contributions. Section III describes the research design and data collection methods employed in our study, providing the foundation for our experimental approach. In section V we detail the innovative aspects of our system, both in hardware and software, showcasing the novel features that distinguish our work from existing solutions. section V offers an in-depth analysis of the communication protocols vital to ensuring system security and efficacy. Following this, section VI outlines the various strategies and techniques utilized to achieve seamless system integration. section VII identifies and discusses the challenges and obstacles encountered in achieving interoperability within the scope of enterprise integration. The critical evaluation of our system is presented in section VIII where we assess the system against predetermined criteria. In section IX we dissect the outcomes of our experimental implementation, providing a comprehensive analysis of the system's performance and integration capabilities. The paper concludes with section X where we summarize our findings and discuss the potential avenues for future research in this domain.

II. BACKGROUND TO RESEARCH

Access control mechanisms play a pivotal role in protecting physical spaces, information, and resources by managing and

restricting entry. Various access control systems have been explored in the literature, including biometric, card-based, keypad, and mobile-based systems. Among these, RFID smartcards have gained prominence as a widely adopted means of authentication in access control systems [1]. A substantial body of research has investigated the use of RFID smartcards for access control (e.g., [2], [3], [4]). These studies have demonstrated the effectiveness of RFID technology in granting access to authorized individuals while restricting unauthorized access. However, the literature has also identified several challenges associated with RFID smartcards, which can compromise the security and reliability of access control systems. For example, several studies have explored the risks of unauthorized access due to stolen, duplicated, or lost cards [5], [6], [7]. These incidents may enable unauthorized individuals to enter restricted areas, posing significant security threats. Research has highlighted the potential for impersonation attacks, in which an individual uses an authorized person's card to access secure locations [8], [9]. In response to these security concerns, some studies have investigated the practical issues arising from the use of conventional smartcards [10], [11], [12]. This research has underscored the security concerns leading to difficulties accessing restricted areas and decreased efficiency and productivity. Given the identified limitations in security and reliability, a growing body of literature has called for further research and development in access control systems [13], [14], [15]). This research has explored more secure and dependable authentication methods, potentially leading to innovative solutions that enhance access control mechanisms. Some research directions include integrating biometric technologies [16], employing cryptographic techniques [17], [18], and examining the potential of multi-factor authentication systems. The literature on access control systems, particularly those utilizing RFID smartcards, has identified both the advantages and challenges of this technology. By developing and implementing advanced authentication methods, researchers aim to address these challenges and improve security and user experience in access control systems.

III. METHODOLOGY

The methodology employed in this study encompasses a comprehensive research design for the collection of primary data and a summative approach for the analysis of qualitative content. The research endeavour revolves around the exploration of diverse solutions while embracing credible, well-referenced facts, devoid of intentions to disprove existing information or endorse robust opinions. By implementing a comprehensive method design, this study adheres to an interpretive paradigm, allowing for the comprehensive understanding and exploration of complex phenomena. This approach is apt for capturing the intricate nuances, rich contextual details, and subjective experiences embedded within the IoT domain. Employing methods such as observations, and document analysis, the primary data collection

endeavours to garner in-depth insights and multiple perspectives. The qualitative contents amassed through data collection undergo a summative approach during the analysis stage. This process entails systematic coding, categorization, and synthesis of data to identify underlying meanings. The objective is to distil the qualitative contents' essential features and salient aspects, leading to a comprehensive overview of the collected data and the derivation of meaningful insights. Utilizing these methodological frameworks allows the study to delve deeply into the subject matter, facilitate practical problem-solving, and yield tangible outcomes with real-world implications. The adoption of qualitative data collection methods and the incorporation of realism as the theoretical perspective serve to enhance the comprehensiveness and validity of the research findings.

IV. COMMUNICATION PROTOCOLS IN ACCESS CONTROL SYSTEMS

A comprehensive examination of communication protocols in access control systems is crucial to understand their effectiveness and security features. These protocols are designed to facilitate secure information exchange between various entities within the system, including users, devices, and servers. This analysis will delve into the key communication protocols utilized in access control systems, highlighting their distinct characteristics and potential vulnerabilities [19].

A. WIEGAND PROTOCOL

The Wiegand protocol has long been perceived as the prevailing standard within the access control domain. The data speed can accelerate up to 35 Mbit/s (within a 10-meter range) or 100 kbit/s (spanning 1200 meters). The protocol lacks encryption and authentication mechanisms, rendering it vulnerable to data interception, tampering, and replay attacks [20].

B. RS485 PROTOCOL

The RS485 protocol serves as a serial communication protocol extensively utilized in many access control systems, encompassing biometric systems and smartcard readers. The security features are contingent upon the higher-level protocols employed. RS485 boasts numerous advantages over other standards, particularly in the context of applications in noisy industrial settings. The design of RS485 is tailored to exhibit noise tolerance and resilience [21]. It allows multiple devices to communicate on the same bus over long distance of about 4000 metres [22].

C. NFC (NEAR FIELD COMMUNICATION)

NFC is a wireless communication protocol that enables contactless communication between devices over short distances, typically within a few centimetres. Frequently employed in access control systems utilizing smartphones as credentials, NFC facilitates secure data exchange via encryption and cryptographic authentication, rendering it suitable for secure access control applications [10], [23].

D. RFID (RADIO FREQUENCY IDENTIFICATION)

RFID is a wireless communication technology utilizing radio frequency for data exchange between a reader and a tag. Employed in diverse access control systems, such as contactless smartcards and key fobs, the security features of RFID-based systems are dependent on the specific RFID technology implemented (e.g., low-frequency, high-frequency, or ultra-high frequency) and the associated cryptographic schemes [2].

E. BLUETOOTH LOW ENERGY (BLE)

BLE is a wireless communication protocol designed for low-power devices, increasingly utilized in access control systems leveraging smartphones as credentials. Supporting secure communication through encryption and authentication, BLE is well-suited for access control applications. Its vulnerability to relay attacks necessitates the implementation of additional security measures. Old legacy access control systems are progressively adopting more secure and versatile communication protocols to enhance overall security and user experience [24].

V. INTEROPERABILITY TECHNIQUES

By employing different techniques and strategies, organizations can ensure seamless integration of various access control systems devices, and components, fostering improved efficiency, collaboration, and overall performance. The interoperability term is defined as several components' ability to exchange and use the exchanged information [25]. Interoperability is a crucial aspect of modern technology, as it enables seamless integration of various solutions and systems, enhancing efficiency, flexibility, and scalability. In this regard, several techniques have been developed to promote interoperability across different systems.

A. STANDARDIZATION

The development and adoption of industry-wide standards are critical for achieving interoperability. Standards define common protocols, data formats, and interfaces, ensuring that different systems can communicate and work together effectively. Examples of standardization organizations include IEEE, IETF, W3C, and ISO.

B. OPEN PROTOCOLS AND DATA FORMATS

Open protocols and data formats enable different systems to communicate and exchange data seamlessly. Developers can ensure their systems can be easily integrated with other solutions by using openly documented and widely accepted communication protocols (e.g., HTTP, MQTT) and data formats (e.g., JSON, XML).

C. APIS (APPLICATION PROGRAMMING INTERFACES)

APIs serve as intermediaries, allowing different software components to interact with one another. By providing well-defined, consistent, documented APIs, system developers can

ensure that other systems can communicate and integrate with their solutions without requiring extensive modifications.

D. MIDDLEWARE

Middleware represents a software stratum situated between disparate systems, components, or applications, functioning to streamline communication and data exchange. By serving as an intermediary, middleware can facilitate translation among diverse communication protocols, data formats, or interfaces, consequently fostering interoperability between systems that might otherwise remain incompatible. This intermediary layer plays a critical role in bridging the gap between heterogeneous systems, promoting seamless integration, and fostering collaboration in a wide range of technological domains.

E. SERVICE-ORIENTED ARCHITECTURE (SOA)

SOA is an architectural pattern that promotes the development of modular, loosely coupled services that can be easily combined and reused across different systems. By designing systems as a collection of interoperable services, developers can achieve greater flexibility and scalability, allowing for seamless integration with other solutions.

F. DATA TRANSFORMATION AND MAPPING

Data transformation and mapping techniques are used to convert data between different formats or structures, enabling interoperability between systems that use different data representations. Data transformation tools and techniques, such as XSLT (for XML data) and ETL (Extract, Transform, Load) processes, can be employed to facilitate data conversion and integration.

VI. INTEROPERABILITY BARRIERS

The researchers examined the impediments and challenges associated with interoperability within the scope of enterprise integration [26]. They identified the following key barriers:

1) HETEROGENEITY

A diverse array of hardware, software, and communication technologies often characterizes enterprise systems, resulting in complications when facilitating seamless communication and data exchange among disparate system components.

2) LEGACY SYSTEMS

Numerous enterprises maintain legacy systems, which were not initially designed for interoperability. Integration of these systems with contemporary technologies and applications proves challenging due to their antiquated architectures, communication protocols, and data formats.

3) LACK OF STANDARDS

The lack of a universally recognized standard that encompasses all communication protocols, data formats, and interfaces can pose challenges to achieving interoperability.

While some standards do exist, they may not cover all aspects of a system or enjoy widespread adoption within the industry.

4) SEMANTIC INCONSISTENCIES

Varied representation and interpretation of data across systems may give rise to semantic discrepancies. Ensuring a mutual comprehension of data exchanged among systems is imperative for attaining interoperability.

5) ORGANIZATIONAL AND CULTURAL BARRIERS

Organizational and cultural aspects can also influence interoperability. Distinct organizations may possess unique processes, structures, and terminologies, engendering information sharing and collaboration difficulties.

6) SECURITY AND PRIVACY CONCERNS

Guaranteeing the security and privacy of data exchanged between systems is essential for achieving interoperability, which may prove challenging due to the diverse security requirements and policies of distinct organizations and systems. To overcome these barriers, the researchers underscore the significance of formulating and adopting standards, utilizing middleware and service-oriented architectures, and employing modeling techniques to comprehensively understand and address the intricacies of enterprise integration and interoperability. Focusing on access control systems these often encompass a wide variety of hardware components, such as card readers, door controllers, biometric scanners, and intrusion detection sensors. These components may utilize distinct communication protocols, making seamless interaction and data exchange difficult to achieve. As a result, the lack of standardized communication protocols for hardware components within access control systems can hinder overall system interoperability. To alleviate this issue, creating and adopting industry-wide communication standards for hardware components within access control systems is necessary. By establishing universally accepted communication protocols, manufacturers and system integrators can ensure that diverse hardware components interact seamlessly, regardless of their specific technology or vendor. This approach promotes greater system flexibility, as organizations can more easily mix and match components from different manufacturers without sacrificing interoperability. Middleware is an intermediary layer that can translate between different communication protocols, data formats, and interfaces, enabling smooth communication between disparate hardware components. Meanwhile, service-oriented architectures promote modularity and loose coupling, allowing for the flexible combination of different services and systems to meet specific security requirements. Employing modeling techniques can assist in understanding and addressing the complexities of enterprise integration and interoperability. By modeling the interactions between various hardware components and communication protocols in access control systems, system designers and integrators can identify potential bottlenecks, inefficiencies, and incompatibilities [25].

VII. PROPOSED INTEROPERABLE ACCESS CONTROL SYSTEM

In this study, the authors present an innovative hardware design for an access control system that incorporates the use of rolling QR codes while maintaining compatibility with existing access control systems based on the widely implemented RS485 Protocol. The proposed design seeks to augment the security and flexibility of contemporary access control frameworks by integrating the benefits of evolving QR code technology without compromising interoperability. The integration of rolling QR codes into access control systems offers several advantages, including heightened security through the continuous generation of unique, time-sensitive codes and the potential for contactless authentication, thereby enhancing user convenience and mitigating concerns related to physical contact. Furthermore, QR codes can be easily distributed and managed through digital means, such as smartphones or other smart devices, reducing the reliance on physical access cards or tokens. By ensuring compatibility with the RS485 Protocol, the proposed hardware design acknowledges the prevalence of this protocol in existing access control systems. It alleviates potential challenges associated with integrating new technologies into legacy systems. The RS485 Protocol has long been a standard for communication between access control devices, such as card readers and door controllers. Consequently, preserving interoperability with this protocol is essential for the successful implementation of the innovative rolling QR code-based access control system. The developed solution highlights the architectural dependency options in fig. 1 and emphasizes interoperability to ensure seamless communication and collaboration across legacy devices and the proposed solution.

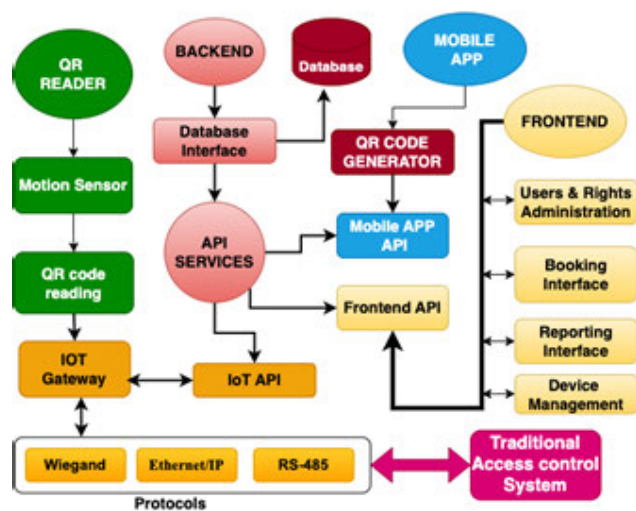


FIGURE 1. Software architecture of the interoperable access control system.

The developed solution aims to tackle the challenges of integrating new systems with existing or legacy devices

while prioritizing interoperability to enable seamless communication and collaboration. The objective was to facilitate smooth interaction and cooperation between a traditional RS485 system and a modern access control system, overcoming potential obstacles arising from differing technologies, data formats, or protocols. The suggested system comprises various components, such as bespoke software and hardware, specifically designed to attain interoperability with the RS485-based access control system. The subsequent sub-points delineate the salient features of the design and implementation procedure.

A. HARDWARE

This layer includes the physical components that enable data collection, input, and communication within the testing environment:

1) TRADITIONAL ACCESS CONTROL SYSTEM EMPLOYING RS485

This system is an electronic security system used to manage access to restricted areas or resources. The key components of the system are:

- The card reader is normally installed at entry points, such as doors. The system used in this experiment is designed to read RFID cards or key fobs, which contain a unique code. When the card is swiped or presented near the reader the code is sent via RS485 protocol to the Access control main unit.
- The access control unit receives the ID data from the card reader and processes it to determine whether access should be granted or denied. The panel stores a database of authorized user IDs and can be programmed with specific access rules, such as time restrictions or limited access to certain areas.
- Upon receiving a signal from the access control main unit, the electric door lock or barrier is either activated or deactivated, allowing or denying access to the user based on their credentials.

Figure 2 illustrates a traditional access control system; in the image, one can observe the card readers and the access control main unit. This image provides a clear overview of the key components that constitute the system. This system was used in our experiment.

B. QR CODE READER

A bespoke circuit board was developed to facilitate the reading of QR codes displayed on mobile devices. Figure 3 depicts the electronic board, which encompasses the requisite components for decoding the QR code and transmitting the information to the IoT gateway. The board employs the GROW GM803 Series Interface Barcode Scanner. The GM803 is equipped with a CMOS sensor featuring a resolution of 640×480 pixels, enabling it to decode 1D barcodes or 2D codes, such as QRCode, DataMatrix, or PDF417. An ESP32 microcontroller [28], featuring a dual-core processor with a clock speed of up to 240 MHz, 520 KB of



FIGURE 2. Traditional access control system.

SRAM, and integrated Wi-Fi and Bluetooth functionalities, is employed for data processing on the QR code reader side. The GM803 scanner transmits the scanned QR code to the ESP32 microcontroller via a serial (RS232) protocol. After the data transmission, the ESP32 microcontroller encrypts the data and forwards it to the IoT gateway using either the integrated 2.4 GHz Wi-Fi protocol or the RS-485 protocol, facilitated by the MAX485, a low-power RS-485 transceiver. The data transmission option can be chosen when setting up the device. A bespoke circuit board was developed to facilitate the reading of QR codes displayed on mobile devices. Figure 3 depicts the electronic board, which encompasses the requisite components for decoding the QR code and transmitting the information to the IoT gateway. The board employs the GROW GM803 Series Interface Barcode Scanner. The GM803 is equipped with a CMOS sensor featuring a resolution of 640×480 pixels, enabling it to decode 1D barcodes or 2D codes, such as QRCode, DataMatrix, or PDF417. This software offers a comprehensive suite of tools and features that enable the efficient and precise creation of electronic circuit board designs. The schematic representation of the designed board, visually illustrating its underlying electrical connections and components, is presented in fig. 4. The data transmission option can be chosen when setting up the device. The design of the board in this study was facilitated by the utilization of Kicad, an open-source software renowned for its capabilities in electronics design and automation. By employing Kicad [29] as the software platform and utilizing the schematic representation, the study ensures a systematic and meticulous approach to board design, incorporating industry-standard practices and facilitating accurate

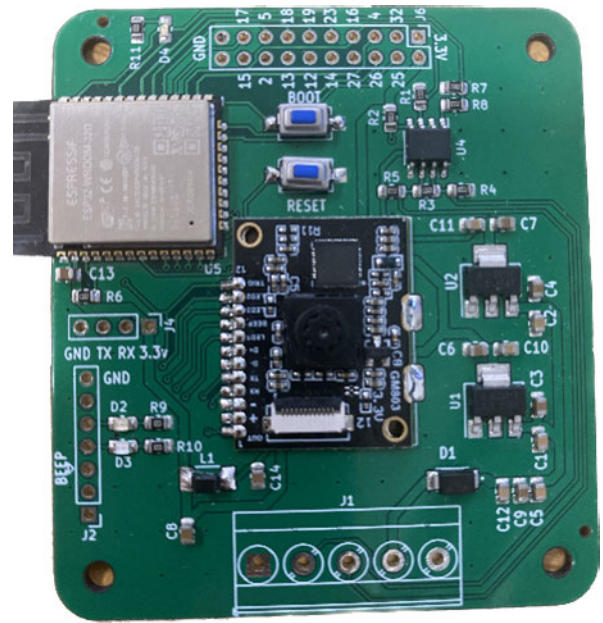
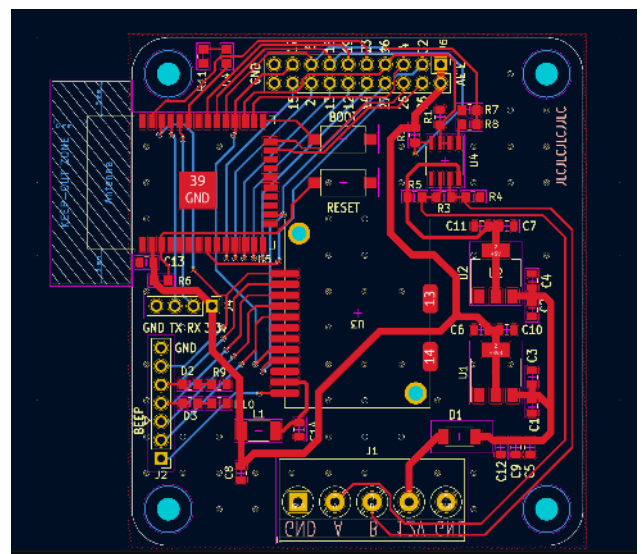


FIGURE 3. QR code reader.



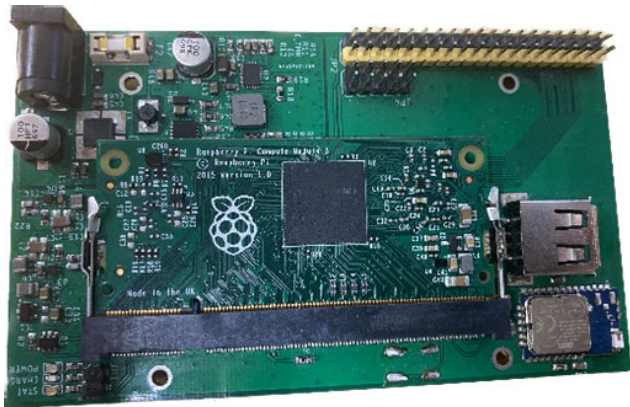


FIGURE 5. The hardware structure of the IoT gateway.

levels to meet the specific requirements of the peripherals while mitigating the power draw directly from the microcontroller. This approach minimizes the potential risks associated with excessive power consumption, safeguarding the microcontroller from issues such as overheating or damage due to high currents.

C. IoT GATEWAY

The IoT gateway was built based on the Compute Module CM3 [30] lite version from the Raspberry Pi Foundation. The CM3 features a Broadcom SoC CPU operating at 1.2 GHz and 1 GB of DDR2 RAM. The IoT gateway board is designed with 20 available GPIOs for a variety of sensors and integrates an MPS 2617 Battery Management chipset. The embedded 1400mAh battery delivers a maximum of 10 hours of battery life at 1000 mA under high load conditions. This duration can be extended by employing a larger battery and adjusting the charging circuit. For connectivity, the device incorporates a Silicon Labs WF111A 802.11 B/G/N Module for 2.4 GHz Wi-Fi and includes an eMMC 8GB 3.3V 200Mhz eMMC 5.0 manufactured by ISSI. The IoT gateway operates on a Debian Linux distribution with a 4.14 kernel version, which is a widely used open-source operating system. Debian is known for its stability, reliability, and extensive software library, making it a popular choice for IoT gateways and embedded systems. In this case, the IoT gateway uses the Linux kernel version 4.14, which introduced several improvements and new features relevant to IoT devices, such as:

- 1) Enhanced support for ARM-based platforms, which are common in IoT devices due to their low power consumption and cost efficiency.
- 2) Improved power management features, enabling IoT devices to operate more efficiently and extend battery life.
- 3) Better security features and bug fixes, ensuring more secure and stable operation for the IoT gateway.

Figure 5 illustrates the arrangement of the components on the circuit board. This visual representation provides a clear overview of the layout and organization of the various

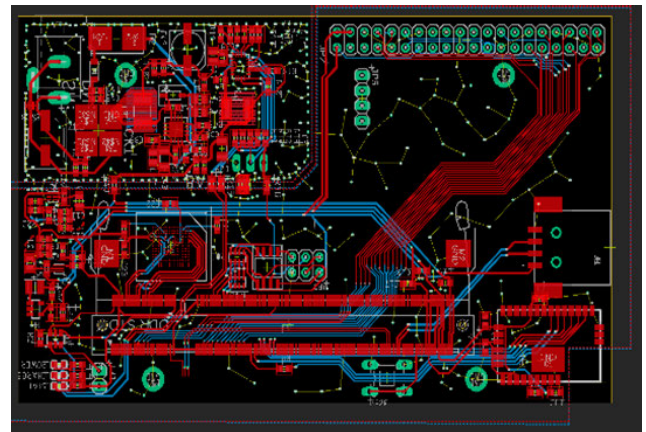


FIGURE 6. Eagle design of the IoT gateway.

elements, including the microcontroller, connectivity modules, and power management components. By examining the components' distribution, we can better understand the design principles, component interactions, and potential areas for optimization or modification. The IoT gateway was designed utilizing Eagle software, a prominent and widely adopted electronic design automation tool renowned for its comprehensive features and capabilities. Eagle software provides a robust and versatile platform that facilitates the precise and efficient creation of complex electronic circuit board designs. Using the advanced functionalities of Eagle, the researchers could navigate the intricacies of the design process and effectively capture the intricate details and specifications of the board. This utilization of industry-standard software and the accompanying schematic visualization facilitates effective communication of the design concept, enhances the reproducibility of the study, and enables other researchers and practitioners to comprehend and evaluate the intricacies of the designed board. At the beginning of the experimental phase, the initial setup encompassed data transmission between the QR code reader and the IOT gateway utilizing the RS232 protocol. However, after this configuration, the research direction shifted its focus toward the utilization of the RS485 protocol for data transmission. Consequently, a supplementary peripheral device, namely a DollaTek serial to RS485 adapter, was introduced and integrated into the existing architecture of the IOT gateway. This adapter, depicted in fig. 7, facilitated the seamless transition from the RS232 protocol to the RS485 protocol, thereby enabling successful and efficient data transmission in the revised experimental scenario. Figure 8 visually portrays the final hardware architecture, illustrating the interconnectedness among the diverse developed components. This comprehensive diagram is a powerful tool for presenting the physical arrangement and connectivity within the system. The hardware architecture encompasses integrating multiple components, each specifically designed to facilitate data transmission between the QR code reader and the IoT gateway. The depiction in Figure 8 aptly showcases

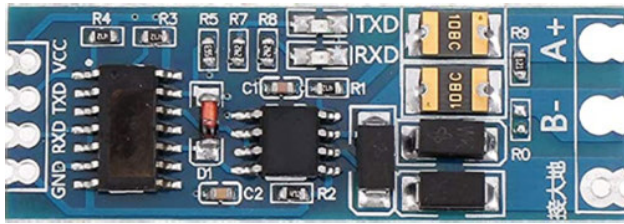


FIGURE 7. SERIAL to RS485.

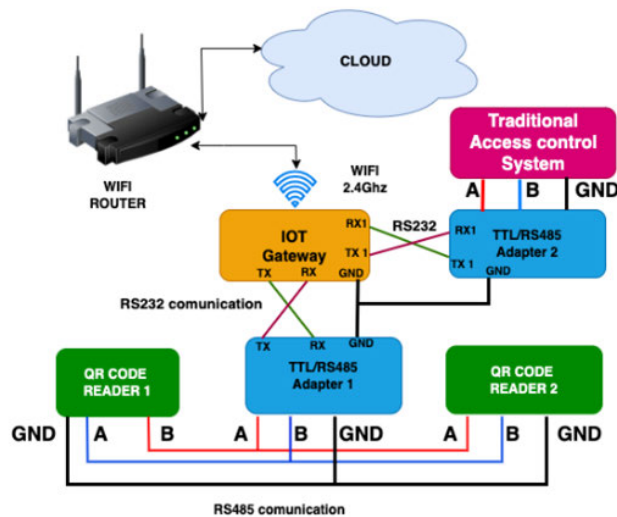


FIGURE 8. Hardware connection framework of the IoT gateway.

the intricate interrelationship and interdependence of these components, collectively constituting a coherent and interconnected system. The figure emphasizes delineating the established connections among the components, thereby outlining the pathways through which data, signals, and power are seamlessly transmitted. This visual representation provides a holistic understanding of the underlying hardware infrastructure, thereby facilitating the analysis and evaluation of the system's overall design. By presenting Figure 8, researchers enable a comprehensive comprehension of the hardware architecture, as it offers a detailed depiction of the physical arrangement and the intricate connectivity within the system. This serves as a valuable resource for analyzing and assessing the system's functionality, allowing for further exploration and potential refinements. This visualization enhances the clarity and transparency of the system, enabling a thorough examination of the physical connectivity and facilitating future analysis, troubleshooting, and potential modifications.

D. SOFTWARE

The implementation of this experiment required the development of various software components that work together to enable efficient communication, data processing, and system management. These software components are described as follows:

1) QR CODE READER EMBEDDED SOFTWARE

The Esp32 microcontrollers are programmed using a language derived from C/C++, facilitating their utilization through various integrated development environments (IDEs), including the widely recognized Arduino IDE and other compatible platforms that support C++ programming and compilation. Different approaches to writing the code were implemented. One of those approaches was EyesWeb [26] open-source platform, which is used to write the block code for the hardware system. But in this particular study, the open-source version of Visual Studio Code, integrated with the platform plugin, was selected as the designated IDE for system development. The adoption of the Open-source version of Visual Studio Code, renowned for its extensive functionality and adaptability, in conjunction with the PlatformIo plugin, significantly contributed to the successful execution of the system development process. This combination of tools provided comprehensive support, encompassing efficient code editing, debugging, and deployment capabilities, thereby optimizing the programming workflow for the Esp32 microcontrollers within the research project. Considering the paramount importance of security in designing the system, all device connections were implemented using the HTTPS protocol to ensure the secure transmission of data to the server. To achieve secure data transmission, the implementation incorporated the WiFi-ClientSecure library, which encrypts the traffic sent to the IoT gateway [27]. This encryption mechanism enhances the confidentiality and integrity of the transmitted data, safeguarding it against unauthorized access and potential tampering. Through the adoption of the HTTPS protocol and the integration of the WiFiClientSecure library, the research project assures the application of robust security measures throughout the data transmission process [28]. This strategic implementation safeguards sensitive information, thereby upholding the system's overall security posture. Figure 9 serves as an exemplification of the implementation of the Visual Studio IDE and the incorporation of the library. The figure showcases the practical utilization of Visual Studio IDE, providing a visual representation of the software environment and the library's integration within the development framework. This illustrative example offers valuable insights into the practical application of the IDE and the library, aiding in the comprehension and replication of the research project's secure data transmission approach.

2) IOT GATEWAY EMBEDDED SOFTWARE

The embedded software component residing on the IoT Gateway plays a crucial role in the system, serving as a fundamental element for facilitating efficient communication and data exchange between the traditional access control system and the newly developed system. This software component plays a significant role in achieving interoperability by translating and unifying disparate protocols utilized by both systems. The protocol unification implemented by the

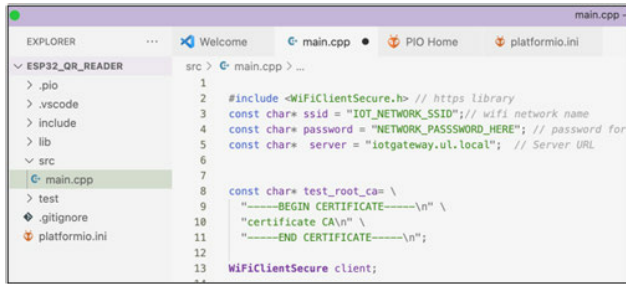


FIGURE 9. Visual studio IDE code for software WiFi initiation.

software allows for seamless communication, enabling real-time insights into the access control process. The software component performs essential functions to process information acquired from the QR code reader, employing specific protocols for data handling. It carries out data cleansing operations to ensure the accuracy and reliability of the obtained information. Subsequently, the software component forwards the processed data to the backend infrastructure using API services. Notably, the communication between the software component and the backend API is conducted using the HTTPS protocol, emphasizing the integration of enhanced security measures [29]. By adopting the HTTPS protocol, the system enhances the confidentiality and integrity of the transmitted data, guarding against potential security threats. This ensures secure and reliable communication between the software component and the backend infrastructure, mitigating the risk of unauthorized access or data breaches. The application residing on the IoT gateway is developed using JavaScript programming language, implementing the Node.js framework. JavaScript is a widely adopted language known for its versatility and extensive support across different platforms. It is particularly well-suited for web development, and with the utilization of the Node.js framework, it extends its capabilities to server-side application development. By employing JavaScript with the Node.js framework, the application benefits from a range of features and libraries that facilitate efficient and scalable server-side programming. Node.js allows for event-driven, non-blocking I/O operations, which enhances the application's performance and responsiveness, making it well-suited for handling concurrent connections and real-time communication. The JavaScript programming language, combined with the Node.js framework, enables the application to seamlessly integrate with other system components. It allows for the implementation of server-side logic, handling data processing, communication with external APIs, and managing the overall functionality of the IoT gateway. The utilization of JavaScript with the Node.js framework empowers the application residing on the IoT gateway with a robust and versatile environment for server-side development. This combination facilitates efficient and scalable programming, ensuring seamless integration within the system architecture and enabling the application to fulfill its intended functionalities effectively. The development of the application

involved the creation of various microservices to cater to multi-protocol testing requirements. These microservices are elucidated as follows:

- 1) **Microservice for RS232/Wiegand/RS485 Protocol Interaction:** This microservice encompasses a distinct software module responsible for reading data received through any of the mentioned protocols. It undertakes data cleansing operations, ensures the security and privacy of exchanged data between systems, and standardizes the received data. Web Service API for WIFI and HTTPS Protocol Data Reception: This microservice focuses on receiving data through the WIFI and HTTPS protocols. It establishes a web service API to facilitate the reception and processing of data obtained via these communication channels.
- 2) **Data Security Microservice:** Dedicated to ensuring data security, this microservice is responsible for handling the encrypted key sent from the mobile app to the QR reader. It decrypts the key using the public and private key keys, extracts the user identification, and verifies the door access against a local database.
- 3) **Hardware Interaction Microservice:** Once the user has been identified and the necessary door access determined, this microservice interacts with the legacy system. It sends a signal to trigger the lock release mechanism, thereby providing access to the designated door.
- 4) **API Microservice:** This microservice operates as a conduit for real-time data updates to about user access events. It consumes an API located in the cloud, allowing for seamless integration of updated information triggered by events. Additionally, this microservice generates logs encompassing user activity and system performance, facilitating subsequent data analysis endeavours. The implementation of these microservices within the application framework enables comprehensive coverage of multi-protocol testing scenarios. Each microservice plays a specific role, contributing to the overall functionality and security of the system. This modular approach streamlines the development process, ensuring efficient communication, data processing, and seamless integration with existing systems.

3) CLOUD SOFTWARE

The Cloud software comprises two main components: the backend and the front end. The backend component plays a vital role in processing requests and managing interactions between the frontend front end and the database. It is developed using Laravel, a widely adopted and powerful PHP web application framework. Laravel follows the Model-View-Controller (MVC) architectural pattern and offers an extensive range of features and tools that empower developers to create intricate, scalable, and high-performing web applications. Laravel encompasses several key features, including a robust routing system, an expressive query builder, a powerful

database abstraction layer with Object-Relational Mapping (ORM) capabilities, user-friendly authentication and authorization mechanisms, caching functionality, and support for multiple file systems. These features collectively enhance the development process and contribute to the creation of sophisticated web applications. The developed application, based on Laravel, incorporates various APIs that facilitate communication and data processing within the SAME (Smart Assistance for Manufacturing Environments) framework. These APIs include:

- **User Management API:** This API utilizes Create, Read, Update, and Delete (CRUD) operations to manage users within the system, enabling effective user administration.
- **IIoT Management API:** This API also leverages CRUD operations to manage Industrial Internet of Things (IIoT) devices and associated data, ensuring efficient management and control of IIoT assets.
- **LOGS API Interface:** The LOGS API Interface is a critical component of our system, providing a conduit for robust data logging and in-depth analysis. This API creates an interface that allows system activities to be tracked, audited, and analyzed effectively, facilitating a better understanding of system behaviors, performance, and potential issues. At its core, the LOGS API Interface serves to capture, store, and retrieve logs and information about the system. This information can include transaction records, user activities, system errors, security incidents, and more. The versatility of data it can handle makes it an invaluable tool for system management and troubleshooting. This API not only logs data but also aids in generating comprehensive reports, delivering insights into the system's operation. These reports can be customized as per the requirement, focusing on areas like user behavior, system performance, error detection, or security auditing.
- **Access Control API:** These services facilitate the management of access control within the system, allowing for the configuration of user access privileges based on designated zones and specific time restrictions. This service enables granular control over user permissions, ensuring that access to various areas or resources is granted or restricted based on predefined criteria related to user identity, designated zones, and time parameters. By utilizing this service, administrators can effectively define and enforce access control policies, optimizing security and maintaining the integrity of the system.
- **Secure Mobile App API Services:** These services serve the purpose of providing secure API endpoints for the mobile app, facilitating the secure transmission of QR codes and ensuring the confidentiality and integrity of the data exchanged. The frontend component of the system encompasses a web client developed using Angular 8, a JavaScript framework built-in TypeScript. This open-source framework offers a comprehensive set of tools and features for efficient web application

FIGURE 10. Access management interface.

development. The frontend system is composed of various interfaces that serve the purpose of facilitating effective system management and providing real-time insights into the access control process. These interfaces empower system administrators with the ability to monitor the status of IoT gateway and QR code readers and perform user management and access control management tasks. Additionally, the interfaces enable administrators to visualize the current state of the overall system. Within the frontend system, a specific interface is dedicated to the setup of user or group access permissions. This interface allows administrators to define access privileges based on location and time parameters. The figure presented below illustrates the aforementioned interface, providing a visual representation of the functionality that enables system administrators to configure access permissions. Through this interface, administrators have granular control over access settings, ensuring that access privileges are aligned with specific locations and timeframes.

4) MOBILE APPLICATION

The mobile application developed plays a pivotal role as a user-accessible platform, enabling them to utilize the secure QR code generation API integrated within the backend system. Additionally, the application encompasses a range of features focused on access control and data analysis, with a specific emphasis on access history tracking. Users are empowered with the functionality to administer their profiles and manage data usage preferences according to their requirements. The development of the mobile application was facilitated by Ionic, an esteemed open-source framework widely recognized for its ability to facilitate the creation of robust and high-performing hybrid applications. By leveraging prominent web technologies such as HTML, CSS, and JavaScript, Ionic seamlessly integrates with widely adopted frameworks like Angular, thereby providing a seamless and



FIGURE 11. Interface of the QR codegeneration within the mobile application.

efficient development environment. In this specific instance, the application interfaces were implemented using Angular 12, enabling the creation of intuitive and user-friendly screens and interactions. Importantly, the Ionic framework ensures broad compatibility with a wide array of mobile devices available on the market, allowing the generated mobile application to be installed and utilized across diverse platforms. Figure 11 provides a visual depiction of the interface specifically dedicated to QR code generation within the mobile application.

VIII. EVALUATION METRICS

Here, the evaluation metrics employed to assess the implemented work are presented with the predetermined requirements for an access control system [30], the following metrics have been identified as essential for conducting a comprehensive evaluation:

- 1) **Security:** The effectiveness of the rolling QR code mechanism in preventing unauthorized access and mitigating the risks associated with code interception or duplication is assessed.
- 2) **Interoperability:** The system's capacity to seamlessly integrate with diverse access control devices, including QR code readers, IoT gateways, and legacy systems, is evaluated. The goal is to ensure compatibility and interoperability among different system components.
- 3) **User Experience:** The user-friendliness and ease of use of the system are considered, encompassing factors such as the simplicity of QR code generation and scanning, intuitive user interfaces, and clear instructions for accessing secured areas.

- 4) **Scalability:** The system's scalability is determined by assessing its ability to handle increasing numbers of users, access points, and simultaneous access requests while maintaining optimal performance and security.
- 5) **Data Privacy:** An evaluation is conducted to assess the system's adherence to data privacy regulations and best practices. This entails verifying that user data and access logs are securely stored, encrypted, and accessible solely to authorized personnel.
- 6) **Auditability:** The system's capability to generate comprehensive access logs and audit trails, enabling traceability and accountability for access control events, is evaluated. The accuracy and completeness of the generated logs are also assessed.
- 7) **Performance:** The system's performance is measured in terms of response time, QR code generation and scanning speed, and overall operational efficiency. This ensures that access control processes are executed promptly and without significant delays.

As given in [31], [32], and [33], these evaluation metrics have been chosen to comprehensively evaluate the access control system based on its predetermined requirements. They provide a robust framework for assessing different aspects of the system's functionality, security, user experience, scalability, data privacy, auditability, and performance.

IX. RESULTS AND ANALYSIS

This section discusses the evaluation results obtained from the design of the system and the experimental implementation conducted in this study. The system underwent various tests to assess its effectiveness and compatibility with traditional legacy systems. Results are presented and analyzed, highlighting the system's performance, functionalities, and integration capabilities. In the used test scenario, the user employs a mobile phone to receive an encrypted key from the server, which is subsequently transformed into a QR code. The user then positions the phone in front of a QR code reader device, which reads the QR code and transmits it to the IoT gateway via RS485, RS232, or Wi-Fi protocols. It is noteworthy that the QR code remains valid for a limited duration of 5 seconds before it regenerates. In this depicted scenario, a mobile phone serves as the primary medium for a user to receive an encrypted key from a server, which is subsequently transformed into a QR code. Once generated, the QR code is then presented to a dedicated QR code reader device. This device, utilizing RS485, RS232, or Wi-Fi protocols, captures the QR code's information and transmits it to an IoT gateway. Upon receiving the QR code information, the IoT gateway undertakes the decryption process, extracting crucial data such as the user's identification and the specific door to be accessed. The decrypted information is verified against a local database to check user access privileges and ensure the QR code hasn't been reused. Access authorization is determined based on the decrypted data, current time, and predefined access criteria. Once access is confirmed, a code is dispatched from the IoT gateway to a traditional access

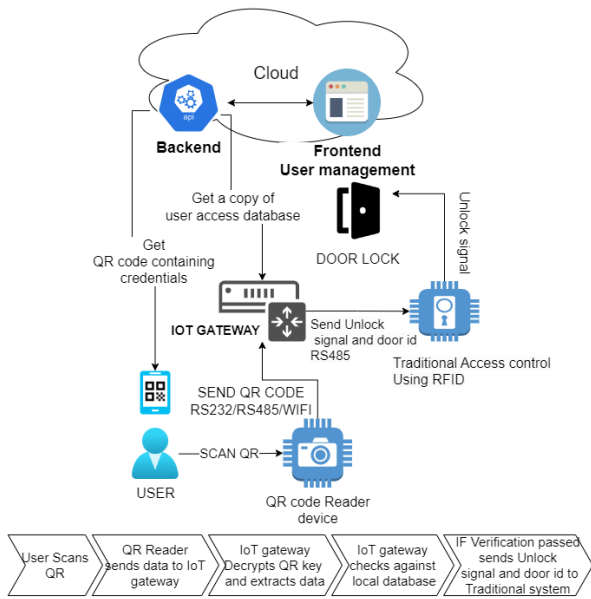


FIGURE 12. Access control process.

control system via the RS485 protocol. This code serves as a signal to activate the lock mechanism, granting entry to the authorized user. Used QR codes are blacklisted and stored in the local database to maintain security and integrity. The local database synchronizes with the backend every minute to keep user information, access times, and authorized doors up-to-date and secure. All system-related information is encrypted with private and public keys to enhance security. This encryption reduces the risk of unauthorized decryption, protecting the confidentiality and integrity of sensitive data throughout the access control process. The outlined process is visually represented in Figure 12 below. The evaluation centered on the effectiveness of the rolling QR code mechanism in preventing unauthorized access and reducing risks of code interception or duplication. System performance was tested using three communication protocols, WiFi, IP, RS232, and RS485, to establish connections between the QR code reader and the IoT gateway. The resulting data, including average response times, incidences of code interception and duplication, and instances of unauthorized access, are presented in Table 1. The results offer valuable insights into the system’s performance, highlighting code interception, duplication, and unauthorized access vulnerabilities. This contributes to a comprehensive evaluation of the rolling QR code mechanism’s effectiveness.

A. SYSTEM RESILIENCE TESTING

The researchers used the Wi-Fi (IP/TCP) protocol to investigate the results. Although the average response time for all protocols was below 100ms, it was observed that the Wi-Fi signal became unstable when the distance between the QR reader and IoT gateway exceeded 50 meters. The researchers found the instability due to the building’s wall

TABLE 1. Comparative analysis of protocols.

Protocol	Average Response time (ms)	Code Interception	Code Duplication	Unauthorized Access
WiFi TCP/IP	32	No	No	No
RS232	18	Yes	Yes	No
RS485	23	Yes	Yes	No

thickness and metal components. Using the HTTPS protocol for data transmission between the QR reader and IoT gateway prevented QR code interception and duplication over Wi-Fi and IP/TCP.

B. SECURITY TESTING

No unauthorized access was observed, indicating the system’s resilience against Wi-Fi-based penetration attempts. One of the main features of HTTPS is that it is encrypted. Wireshark was used to emulate a security attack and capture network traffic. Since HTTPS is encrypted, there’s no way to read it in Wireshark. The encrypted data was decrypted using the cipher negotiated by the TLS protocol of HTTPS. The decrypted data was found to be unaltered and the integrity of the data was proven by the hash/compression algorithm.

For the RS232 protocol, the researchers found that the response time was comparable to Wi-Fi, under 100ms. However, it was discovered that the data sent in clear text and lack of encryption made it susceptible to interception and duplication. Interception and reading of QR codes were possible by connecting a USB to the RS232 adapter to the QR code reader board. However, unauthorized access was prevented due to the expiration time and blacklisting of used QR codes. Similar tests with the RS485 protocol showed the same results, with no unauthorized access observed. Furthermore, the researchers calculated no data losses using a 30-meter cable with the RS485 protocol. The results of this study indicate that the rolling QR code mechanism is highly effective in thwarting unauthorized access attempts. Despite the possibility of QR code interception with RS232 and RS485 protocols, the use of expiration times and blacklisting effectively prevented unauthorized access, thereby enhancing the system’s overall security.

C. SCALABILITY TESTING

The system’s scalability was thoroughly evaluated, demonstrating its ability to handle increasing users, access points, and simultaneous access requests while maintaining optimal performance and security. Extensive testing showed that the system efficiently managed a substantial workload without compromising service quality. For example, a single IoT gateway efficiently managed 45 QR code readers during testing without degrading service performance, highlighting the system’s robust scalability. The gateway can manage between 100-1000 concurrent connections when operating

solely with the local database. However, when the same environment is simulated with cloud database synchronization, the gateway can handle only 100-500 concurrent connections before experiencing any performance degradation. Additionally, the system's seamless integration with various legacy access control devices was evaluated using the RS485 communication system. As given in fig. 13, the proposed system efficiently exchanged data between the IoT gateway and traditional access control systems without complex installations, maintaining a secure connection. A user study with 25 participants assessed the system's user-friendliness, focusing on the simplicity of QR code generation and scanning and the intuitiveness of user interfaces.

D. GDPR COMPLIANCE

An assessment was conducted to ensure the system's compliance with GDPR requirements. This included verifying that user data and access logs were securely stored, encrypted, and accessible only to authorized individuals. Personal information was encrypted before storage using an encryption key derived from hardware IDs associated with the IoT gateway. To decrypt and access personal information that could identify the user, the IoT gateway must be online and accessible to provide the necessary key via backend APIs. Therefore, decrypting data becomes highly improbable without a physical IoT gateway during a security breach. Additionally, the QR codes generated by the system do not contain any personal information about the user, further safeguarding data privacy. Implementing these measures ensures GDPR compliance through robust encryption, reliance on the IoT gateway for decryption, and exclusion of personal details in QR codes, maintaining high data privacy and security. The evaluation assessed the system's ability to generate comprehensive access logs and audit trails for traceability and accountability. Specific measures within the IoT gateway software enabled accurate logging of access control events by storing all requests from the QR reader to the IoT gateway in the cloud database via an API, ensuring seamless data transmission and logging. A dedicated listener monitored the RS485 connection between the IoT gateway and the traditional access control system, capturing and forwarding all data exchanges to the cloud server via APIs. The accuracy of the access logs was rigorously evaluated, with no data loss detected. All data exchanges were effectively recorded and securely stored in the cloud database.

E. OVERALL SYSTEM PERFORMANCE

The system demonstrated its ability to generate comprehensive access logs and audit trails, ensuring data accuracy and integrity, thus enhancing traceability and accountability for access control events. The system was evaluated over three months, operating seamlessly without manual intervention. This assessment confirmed the system's ability to consistently deliver its intended functionality and maintain optimal performance, demonstrating robustness and stability in autonomous operation.



FIGURE 13. Framework for access control.

F. USE CASE SCENARIOS

The proposed system can have various real-time applications.

- **Access Management:** Rolling QR codes improves efficiency by providing temporary access while maintaining a robust log for audit purposes.
- **User convenience:** Rolling QR code system eliminates physical keys or cards, enhancing security against theft or duplication. The contactless solution also minimizes logistical challenges.
- **Compatibility and functionality:** Ease of integration legacy systems ensures compatibility and adding extra capabilities without replacing existing infrastructure, saving costs.
- **Scalability:** Rolling QR codes is scalable and supports multiple users and devices without compromising security, ideal for large-scale industrial operations.

G. TRADE OFFS OF USING QR CODES

Frequent QR code regeneration offers enhanced security benefits but also has trade-offs. The following list presents some key considerations:

- **Increased Data Transmission:** Regular updates of QR codes can prevent unauthorized access by ensuring that each code is unique and time-sensitive. However, this requires frequent data transmission between the server and the device, increasing the network traffic and slowing operations.
- **Processing Overhead:** Constant generation and validation of new QR codes leads to higher computational costs and potential delays.
- **User Convenience:** Users may find it inconvenient to scan new QR codes frequently, especially if the process is slow or the users need to update their app or device settings.

X. CONCLUSION

As described in sections V to VII, this study has successfully demonstrated the interoperability of the proposed access control system with legacy systems through a comprehensive assessment of the system's functionality, performance, security, scalability, user experience, data privacy,

auditability, and communication protocols, with a primary focus on achieving interoperability. Figure 13 depicts the overall framework designed in this study.

The evaluation confirmed that the proposed system effectively communicated with old legacy systems, establishing seamless integration and interoperability between the components. The system's functionality was thoroughly evaluated, ensuring it met the requirements and specifications for effective communication with legacy systems. Performance tests assessed the system's responsiveness, efficiency, and reliability in facilitating communication and data exchange with the legacy systems. Moreover, the study paid significant attention to the system's security aspects, evaluating its ability to maintain data integrity and confidentiality during interactions with legacy systems. Robust security measures, such as encryption protocols and access controls, were implemented to safeguard sensitive information and prevent unauthorized access. Scalability, another critical aspect of the evaluation, examined the system's ability to handle increasing workloads and accommodate a growing number of users and access points while maintaining optimal performance. The system demonstrated its scalability by effectively scaling up and meeting the demands of legacy systems without compromising functionality or performance. The evaluation also focused on the user experience, assessing the system's ease of use, intuitiveness, and overall user satisfaction when interacting with legacy systems. The study focused on designing user-friendly interfaces and streamlining processes to ensure a seamless and positive user experience. Data privacy and auditability were paramount in the evaluation, ensuring the system complied with relevant regulations and best practices. Measures were taken to protect user data, secure access logs, and maintain an auditable trail of system activities with legacy systems. We thoroughly examined the communication protocols employed by the system, assessing their compatibility with legacy systems and ability to facilitate effective and reliable communication. The evaluation affirmed the proposed access control system's interoperability with old legacy systems, highlighting its functionality, performance, security, scalability, user experience, data privacy, auditability, and communication protocol compatibility. The findings of this study contribute to the body of knowledge surrounding access control systems and provide insights into the development of interoperable solutions for integrating with legacy systems. Figure 13 depicts key findings highlighted in this study.

The 'Results and Analysis' section of the study thoroughly examined the system's proficiency in generating detailed access logs and comprehensive audit trails. Our experimental results demonstrate that the system not only records each access event with precision but also captures a wealth of data associated with these events. This includes the time stamp, user identification, access point details, and the authentication method. The system's capability to log such granular details is pivotal in enhancing security and monitoring, as it allows for meticulous tracking of access patterns and

identification of potential security breaches. Our analysis shows that the system's audit trails are robust and comprehensive, providing a detailed chronological record of all activities and changes, including administrative actions. These audit trails are essential for maintaining system integrity, offering transparency and accountability crucial for security audits and regulatory compliance. The system's advanced data analytics tools extract meaningful insights from access logs and audit trails, helping administrators identify trends, detect anomalies, and make informed decisions about security policies and system enhancements. Its reporting capabilities allow for customized reports, providing valuable data for security analysis and organizational planning. Our results indicate the system's capabilities in generating and analyzing access logs and audit trails are comprehensive and highly efficient. Integrating IoT technology with advanced data processing ensures that the system can handle large volumes of data seamlessly, maintaining high performance even under heavy usage. This efficiency is a testament to the system's robust design and potential to be a reliable and effective tool in various security-sensitive environments.

While HTTPS offers a secure communication channel, it is still vulnerable to attacks like Man-in-the-Middle (MitM) and SSL/TLS vulnerabilities. Although our research does not specifically test these attacks and vulnerabilities, future work could explore these possibilities. Investigating the combination of multiple authentication methods, such as rolling QR codes with NFC, could provide a robust and secure solution which will be looked in future studies.

DECLARATION OF COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships.

STATEMENT

During the preparation of this work, the author utilized Grammarly and other spelling services to enhance the grammatical accuracy and readability of the paper. Following the utilization of this tools/services, the author carefully reviewed and edited the content, taking full responsibility for the final publication.

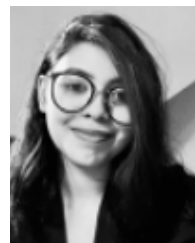
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